

Clase Song

```
public class Song {  
    private String title;  
    private double duration;  
    public Song(String title, double duration) {  
        this.title = title;  
        this.duration = duration;  
    }  
    public String getTitle() {  
        return title;  
    }  
    @Override  
    public String toString() {  
        return title + ": " + duration;  
    }  
}
```

Clase Album

```
import java.util.ArrayList;  
import java.util.LinkedList;  
public class Album {  
    private String nombre;  
    private String artista;  
    private ArrayList<Song> canciones;  
    public Album(String nombre, String artista) {  
        this.nombre = nombre;  
        this.artista = artista;  
        canciones = new ArrayList<>();  
    }  
    public boolean addSong(String title, double duration) {
```

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    if (findSong(title) == null) {
        canciones.add(new Song(title, duration));
        return true;
    }
    return false;
}

private Song findSong(String title) {
    for (Song song : canciones) {
        if (song.getTitle().equalsIgnoreCase(title)) {
            return song;
        }
    }
    return null;
}

public boolean addToPlayList(int trackNumber, LinkedList<Song> playList) {
    int index = trackNumber - 1;
    if (index >= 0 && index < canciones.size()) {
        playList.add(canciones.get(index));
        return true;
    }
    return false;
}

public boolean addToPlayList(String title, LinkedList<Song> playList) {
    Song song = findSong(title);
    if (song != null) {
        playList.add(song);
        return true;
    }
    return false;
}

```

```
}  
}
```

Clase Main

```
import java.util.ArrayList;  
import java.util.LinkedList;  
import java.util.ListIterator;  
import java.util.Scanner;  
  
public class Main {  
    private static Scanner scanner = new Scanner(System.in);  
    public static void main(String[] args) {  
        ArrayList<Album> albums = new ArrayList<>();  
        // Álbum 1  
        Album album1 = new Album("Divide", "Ed Sheeran");  
        album1.addSong("Shape of You", 4.24);  
        album1.addSong("Perfect", 4.40);  
        album1.addSong("Galway Girl", 2.50);  
        albums.add(album1);  
        // Álbum 2  
        Album album2 = new Album("Thriller", "Michael Jackson");  
        album2.addSong("Thriller", 5.57);  
        album2.addSong("Beat It", 4.18);  
        album2.addSong("Billie Jean", 4.54);  
        albums.add(album2);  
        // Playlist  
        LinkedList<Song> playList = new LinkedList<>();  
        album1.addToPlayList("Shape of You", playList);  
        album1.addToPlayList(2, playList);  
        album2.addToPlayList("Thriller", playList);  
        album2.addToPlayList(3, playList);  
    }  
}
```

```

        printList(playList);
        play(playList);
    }

    public static void printMenu() {
        System.out.println("\nOpciones:");
        System.out.println("0 - Salir");
        System.out.println("1 - Siguiente canción");
        System.out.println("2 - Canción anterior");
        System.out.println("3 - Repetir canción actual");
        System.out.println("4 - Mostrar playlist");
        System.out.println("5 - Mostrar menú");
        System.out.println("6 - Eliminar canción actual");
    }

    public static void printList(LinkedList<Song> playList) {
        ListIterator<Song> iterator = playList.listIterator();
        System.out.println("\nPlaylist:");
        while (iterator.hasNext()) {
            System.out.println(iterator.next());
        }
    }

    public static void play(LinkedList<Song> playList) {
        Scanner scanner = new Scanner(System.in);
        boolean quit = false;
        boolean goingForward = true;
        ListIterator<Song> iterator = playList.listIterator();
        if (playList.size() == 0) {
            System.out.println("No hay canciones en la playlist.");
            return;
        } else {

```

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        System.out.println("Reproduciendo " + iterator.next().toString());

        printMenu();
    }

    while (!quit) {

        int action = scanner.nextInt();

        switch (action) {

            case 0:

                System.out.println("Saliendo de la playlist.");

                quit = true;

                break;

            case 1:

                if (!goingForward) {

                    if (iterator.hasNext()) {

                        iterator.next();

                    }

                    goingForward = true;

                }

                if (iterator.hasNext()) {

                    System.out.println("Reproduciendo " + iterator.next().toString());

                } else {

                    System.out.println("Has llegado al final.");

                    goingForward = false;

                }

                break;

            case 2:

                if (goingForward) {

                    if (iterator.hasPrevious()) {

                        iterator.previous();

                    }

                }

            }
        }
    }
}

```

```

    }

    goingForward = false;
}

if (iterator.hasPrevious()) {
    System.out.println("Reproduciendo " + iterator.previous().toString());
} else {
    System.out.println("Estamos al inicio de la playlist.");
    goingForward = true;
}

break;

case 3:

    if (goingForward) {
        if (iterator.hasPrevious()) {
            System.out.println("Repitiendo " + iterator.previous().toString());
            goingForward = false;
        }
    } else {
        if (iterator.hasNext()) {
            System.out.println("Repitiendo " + iterator.next().toString());
            goingForward = true;
        }
    }

    break;

case 4:

    printList(playList);

    break;

case 5:

    printMenu();

    break;

```

case 6:

```
    if (playList.size() > 0) {  
        iterator.remove();  
        if (iterator.hasNext()) {  
            System.out.println("Ahora suena " + iterator.next());  
        } else if (iterator.hasPrevious()) {  
            System.out.println("Ahora suena " + iterator.previous());  
        }  
    }  
    break;  
}  
}  
}  
}
```